



The rise of the research automaton: science as process or product in the era of generative AI?

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Abstract

Generative Artificial Intelligence (Gen AI) now allows for the seeming automation of most if not all steps in the scientific research lifecycle, giving rise to what I refer to as the *Research Automaton* – the production of science-like output with minimal meaningful human engagement. This development is often framed through a techno-solutionist lens, promising efficiency gains by treating the traditional, often strenuous, research process as a problem to be solved. This paper challenges that perspective, arguing that the intrinsic value of science lies in this very *process* rather than solely in the *product*. Uncritically embracing automation thus entails eroding the formative experiences crucial for researcher development, particularly for early-career researchers, leading to potential skill atrophy and undermining the long-term innovative capacity of science. Drawing on both normative arguments about science as a vocation and pragmatic concerns about preserving essential cognitive and critical skills, I advocate for resisting the Research Automaton, while acknowledging the potential for AI to *augment* human capabilities when used judiciously within a hybrid cognitive constellation. I conclude by outlining practical implications for researchers, supervisors, institutions, policymakers, and publishers in navigating the integration of Gen AI in research.

Keywords Generative AI · Research automation · Scientific process · Techno-solutionism · Ethics of AI

1 Introduction

Imagine being able to produce a scientific paper of acceptable, or even *high*, quality without reading any articles, books, or other scientific material in the process. Without analyzing data or writing it up. Without even coming up with the idea for the paper yourself. If scientific papers themselves are considered intrinsically valuable, this would surely be a good thing. But are they? Here I argue that we should not assign intrinsic value to scientific *output*; scientific output is primarily valuable because it is the result of a *process* that is deeply meaningful for individuals and important for our societies.

Today, however, this process perspective is fundamentally challenged by voices proposing that we use artificial intelligence (AI) to automate science and “fix” the perceived problem of traditional science requiring much time, human resource, and at times both pain and suffering. We thus have a situation in which AI evangelists, and others keen on rationalizing academia, reframe the traditional process of research as a problem in need of a solution. The messy nature of research is seen as amenable to human rationality, and with an engineering mindset, they set out to reform academia by technological optimisation and automation. The push for more AI in higher education and research thus represents a typical case of techno-solutionism (Sætra and Selinger 2024). While such “solutions” might indeed generate more output, it would also entail reframing and radically altering the nature of academia and the life and role of researchers.

I challenge these voices as I emphasize how the increasing automation of the scientific process is likely to be detrimental for researchers, science, and society. The sometimes strenuous, slow, and frustrating tasks involved in manually getting an overview of the existing literature in

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a field, understanding theories, methods and how others solve challenges, reading and coding interview notes or analyzing data, and processing this information into scientific output are arguably the key formative elements in becoming, and being, a researcher or scientist.

If my position as stated here has merit, scientific output has an instrumental value as the *byproduct* of intrinsically valuable steps in the scientific process. And if so, the uncritical automation of these tasks is a risky endeavor for academics, academia, and the societies in which academia is embedded.

In this article, I hone in on how generative AI (Gen AI) (Sætra 2023) enables the rise of *the Research Automaton*, the automated researcher that allows for the production of science-like output without any specific human being meaningfully involved in a traditional research process. I identify both normative and pragmatic reasons to resist this automaton. First, through the notion of science as a vocation (Weber 1946), I identify a set of aspects of doing research that are valuable both for those doing it and for society at large. Second, because there is an important formative and developmental aspect of doing science, automating research might produce short-term output efficiencies at the cost of a long-term collapse due to researchers no longer being capable of driving knowledge production beyond new frontiers and thus no longer provide new data to improve Gen AI models.

This article begins with a brief examination of the linkages between technology and research. I then proceed to present the current situation where Gen AI has emerged as a technology with the potential to radically disrupt science and academia, and move on to a discussion of how and why we should, and perhaps at times shouldn't, be concerned about the rise of the Research Automaton, before the final section where I discuss the implications of the analysis for key stakeholders involved in research, with a particular focus on academia.

2 The entwined histories of science and technology

AI's capabilities for information processing, modeling, and content generation are remarkable, raising important questions regarding the role of the human in the scientific process (Dwivedi et al. 2023; Hassabis & Manyika 2024; Sætra 2018). However, before starting answering such questions, we must understand both the long relationship of technology and science and the propositions that are currently being made about the potential for AI to revolutionize science and humans' role in it.

2.1 The researcher as a tool user

The histories of science and technology are intimately interconnected. According to Simon (2008), *science* is humanity's quest to understand the world through knowledge, while *technology* possesses a different, unique quality. It concerns the world humans want to *create*: the use of instruments to transform the world as it *is* into the world that *ought to be*. However, changes in technology have significant effects on science, breaking down barriers and generating new territories for research. New technology expands our access to the world, so to speak (Godfrey-Smith 2003). The reverse process is equally as important, where the production of knowledge shapes what is possible for the design and engineering of technology.

Humans are, in essence, a creative and tool-using species (Burry 2005). This goes for all aspects of human life, including our search for food, shelter, mates, power, and *knowledge*. As new tools have emerged, these have often transformed the limits of research. In some instances, new technologies have upended scientific debates.

An early modern example is the air pump, a peculiar instrument that resolved a long-running dispute between the philosophers and scientists Thomas Hobbes and Robert Boyle (Shapin and Schaffer 2011). It allowed for rigorous experimentation and empirical evidence, and through that Boyle's theories won out over Hobbes's speculations, ending a long-standing debate and making question regarding *vacuum* one of *evidence* and not speculation and argument. A similar tendency is seen with the emergence of big data and AI as seen in, for example, the proposition that we no longer need theory and human speculation because "with enough data, the numbers speak for themselves" (Anderson 2008). While Gen AI systems might appear to deal with language, text, images, etc., they *are* based on quantitative data, numbers, and thus represents a new stage in the quantification of the sciences (Di Fiore et al. 2023).

The idea that new technologies are reshaping the nature and scope of research is nothing new. In the fifteenth century, the printing press revolutionized the distribution of texts and ideas (Eisenstein 2013). The enhanced scale of information transfer transformed culture and society by enabling broad discussion among the population, which some argue was essential to the formation of the European sciences. However, developments related to reading and writing have always been coupled with the moral panic of elites (Furedi 2016), which is a good reason to also be skeptical of my warnings about Gen AI, perhaps. The spread of writing was perceived as deeply dangerous by Plato and his ilk, and there was no shortage of concern about the moral consequences of the printing press and mass dissemination of the printed word.

Social and economic relations are often reworked as technologies and knowledge evolve. For example, agricultural and military tools and expertise were essential for the scaling of human populations (Scott 2017; Turchin et al. 2022), and the mass deployment of factory machinery enabled unforeseen levels of commodity production (Thompson 2013). The question I focus on in this article is how technology also changes *science* and *research* in similar ways. Are efficiency and rational organization, two things that technology, according to critics like Jacques Ellul (1964), tends to promote, applicable also to science, making things like the aforementioned “scaling” and “productivity” goals to be optimized through technology use in research as well?

2.2 Research and the rise of artificial intelligence

Debates on the automation of the research process have a long history. In the early formation of cybernetics during the 1950s, Norbert Wiener (1989) considered how the interplay of humans and machines would redefine the use of information, dramatically restructuring the organization of societies. Such discussions proceed to this day, where philosophers continue to speculate on the way information processing systems are reshaping the form and structure of human thought (Hui, 2024). Partly by allowing us to automate and optimize tasks, and partly by allowing us to do brand new things that were impossible without these technologies.

Technological changes have emerged throughout history alongside considerable shifts in scientific practice and social relations. With each turn, new forms of knowledge are produced, while entirely new spaces of meaning are generated. At the same time, prevailing scientific paradigms and cultural norms are lost (Kuhn 2009). Technological and scientific changes function as a mode of creative destruction, acting as both productive and destructive forces (Schumpeter 2013). Today, AI has the power to significantly alter scientific research. It is arguably already doing so, for better or worse. As OpenAI’s model *o1* was released in ChatGPT (OpenAI 2024), for example, *Nature* reported that scientists were “in awe” of this new system that purportedly “excels at science” and beats PhD scholars in science tests (Jones 2024). While the scope of the breakthroughs is debatable, it is clear that many perceive these models as groundbreaking and revolutionary tools for research.

Also, with Google DeepMind’s AlphaFold (Jumper et al. 2021), the boundaries of biology were and are pushed, and AI profiles Hassabis and Manyika (2024) argue that AI will help us “understand the very fabric of reality”. The AlphaFold model has produced a database of 200 million protein structures, which without AI, would have taken decades, perhaps longer, to compose.

While the AI hype is strong (Funk 2024), there is certainly also real potential for change inherent in AI. In 2024,

two Nobel Prizes, in physics and chemistry, were awarded to people involved in the development and use of methods and systems that are the basis of Gen AI, demonstrating the significant impact this technology now has in research (Cristianini 2024). One of these went to Google DeepMind founders for their work on AlphaFold. The enthusiasm in both the AI community and parts of the scientific community is high as science is seen as an exciting new frontier for AI, while AI provides new and exciting instruments for scientists (Ioannidis 2024). In the world of corporate research and development, AI is also heavily used, as for example seen in one study that showed how AI led to significant productivity gains in innovation capacity in materials discovery as measured by patent filings and downstream product innovation (Toner-Rodgers 2024).

However impressive, we must note that problems, such as folding problem and material discovery, could be categorized as “easy problems” where what we want to find out is quite clear; it is as of yet harder to argue that current AI systems can solve the “hard problems” which revolves around problem *formulation* and framing of research (Battleday & Gershman 2024). Furthermore, as Mohammed AlQuraishi aptly asks in a story about the unfolding [sic] of the folding problem and AI saga: “Are we even legitimately doing science anymore, or is it something different?” (McDonough 2024), and proceeds to state:

In the past, what drove us was understanding. With this wave of AI, we’re maybe losing some of that. We’re able to get the practical benefits, but we’re not necessarily gaining intellectual benefits (McDonough 2024).

3 Generative AI and the rise of the research automaton

To illustrate the emergence of the hypothesized Research Automaton, I have developed the following brief plausible near-future scenario.

Sarah took a sip from her coffee. Almost cold by the time she had gotten to her cubicle. Better cold good coffee than the sludge they have in the office, she thought. 8 am, just on time. She had a look at today’s schedule. Her supervisor had outlined eight products today. She removed her scarf, fired up the computer and got going. “Geoengineering and ethics” was the first one. She loaded the gap-finding module and crafted the prompt, and the system immediately suggested gaps in the existing literature and a set of proposed research questions. She checked the assignment again. This was a standard product, so she would not need new data here as this would be a theoretical review article. She clicked the literature review but-

ton. Another sip from the coffee cup, and it was done. “Identify and describe methods” was the next step. Another click of the button and the methods section was all done. She had a quick glance at the result. Nice, she thought, this would be a philosophical investigation using counterfactual analysis to explore the ethical implications of cloud brightening. Had she had the time, she’d love to give it a read. She had no such thing. The discussion module was loaded, running, and soon complete. She hit the finalize button, and the introduction, conclusion, and literature lists were done. Nice and smooth, she thought, just as she had run out of coffee. Now she just had to get the product out there. She checked the boxes for the three key journals suggested for this article, and then ticked the boxes for an op-ed, LinkedIn, and Facebook posts. Click. Done. The product was complete – manuscripts and title pages for the three journals and the other material. Zipped it and sent it to her supervisor. Science is quite fun, she thought. And easy. Can’t imagine what it was like back in the 2010’s.

Modern researchers might not work in cubicles and have schedules, managers, supervisors, or sludge for coffee. Or wait, many actually do. Restart: modern researchers in general are, of course, not working along the lines imagined in the vignette above. However, the story is not as far-fetched as you might think. New PhD students and early-career researchers have used Gen AI tools in school already, and they are increasingly entering academic institutions and communities where they are immediately introduced to a wide range of tools resembling bits and pieces of what is hinted at in the scenario above. Furthermore, there are people working in so-called “paper mills” that do approximately what Sarah was described as doing, for others and for pay (Prez-Neri et al. 2022).

The technological phenomenon we encounter here is radically different from the use of technological instruments and machines described in the previous section. Rather than *augmenting* human senses and cognition, allowing the researcher to do more, identify and measure new things, calculate and test models they could not do manually, the vignette describes a situation in which research production

is *automated*. Much like the transition from work by hand and traditional craftsmanship to roboticized mass production, the role of the human worker is drastically changed, if not wholly eliminated. However, while human labor appears to be gone when Gen AI “works”, it is imperative that we always bring forth both how these systems have, often illegitimately, appropriated historical human labor and that they are completely dependent on the ongoing, and often exploitative, human labor performed by data workers (DAIR Institute 2025).

Sarah illustrates what I call the *rise of the Research Automaton*. It is now clear that such a system is technologically conceivable, and some have even presented more formalized and serious attempts to develop such automatons, such as the “AI Scientist” that is argued to be able to both conduct and disseminate scientific research autonomously (Lu et al. 2024).

The Research Automaton is at the extreme end of a continuum of human–machine constellations. On the other end, we have science based solely on human cognition, then humans using advanced tools such as computers, then more comprehensive human augmentation and collective intelligence constellations, and finally the Research Automaton (Fig. 1).

The key technology driving the Research Automaton is Gen AI (Sætra 2023). Generative AI refers to a class of machine learning models designed to produce new content, such as text, images, audio, or code, based on training data. These models are often based on transformer architectures, a form of deep learning neural networks. When embedded within systems such as ChatGPT, they allow users to interactively generate various types of content. Initially, much attention was focused on large language models (LLMs) like OpenAI’s GPT series, which generate text. However, as generative systems now also include models that produce images, video, and audio, such as diffusion models, the term “generative AI” has become an umbrella label for a variety of content-generating technologies beyond just LLMs.

ChatGPT initiated the breakthrough of Gen AI in non-specialist circles. Launched by OpenAI late 2022, this service allowed everyone to interact with OpenAI’s GPT-3 model. And interact with it people did. Young and old, students and professionals, all found the system both intuitive,

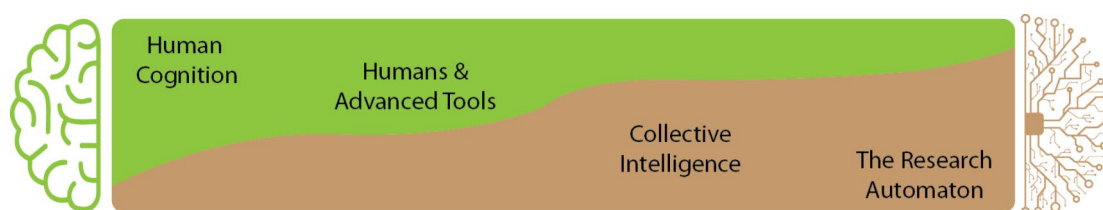


Fig. 1 The continuum from human cognition only to the Research Automaton

fun, and useful. Previous generations of LLMs were already potentially useful but coupled with the radical accessibility and ease of use that came with ChatGPT’s interface and free-to-use model, the use of the technology exploded, in line with established theories of technology acceptance (Davis 1989).

Scientists and higher education students at the PhD level were, and are, no exceptions. Much proverbial ink has been spilled on the benefits and harms of Gen AI, why we should not use it, how to use it more effectively, and how to use it responsibly (Dwivedi et al. 2023; Kenthapadi et al. 2023; Watermeyer et al. 2024). Regardless of the academic musings over what Gen AI could and should be, people immediately put it to work. In the words of William Gibson, the street finds its own use for things. Regardless of what OpenAI originally intended, their solutions now permeate the research field.

While ChatGPT is a sort of generalist, there are many competing generalists, but also specialist systems targeted directly at doing different parts of the work Sarah did in the vignette. An illustrative and non-exhaustive list of some Gen AI tools used for research purposes is provided in Table 1.

In the somewhat early days of Gen AI, we got an excellent example of Big Tech’s hubris and attitude to science and research. Meta introduced their *Galactica* LLM (Taylor et al. 2022). Trained on “48 million papers, textbooks and lecture notes, millions of compounds and proteins, scientific websites, encyclopedias and more” (Taylor et al. 2022), it was supposed to allow us all to harness the wealth of human scientific labors easily and effortlessly.

Table 1 Example Gen AI tools useful for or directly developed for use in research

Field or activity	Example Tools
Generalist Systems	ChatGPT , Claude , Gemini , Mistral , DeepSeek
Literature Review	Elicit , scite.ai , ResearchRabbit
Academic Writing	GrammarlyGO , Writefull , Paperpal
Data Analysis	ChatGPT Advanced Data Analysis , NVivo
Transcription	Whisper , Otter.ai

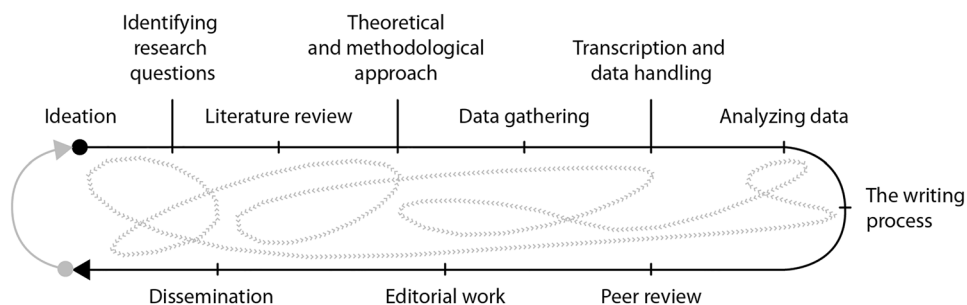
But users immediately found that the model “hallucinated” and made up both “facts” and non-existing scientific references. The model was laughed off the internet in just a couple of days, never to be seen again (Chartier-Edwards et al. 2024). However, in 2025, we have new models and systems, and with services such as Open AI’s “deep research” (OpenAI 2025), we are once again seeing social media sites and hype media filled with the musings of heralds of the demise of the human researcher.

Serious companies targeting academic institutions are pitching their systems with increased efficiency, ease, and productivity as the key selling points. Like they’re selling some *fix* or *solution* to the problem of having to do research to produce research. One popular service states, no kidding, that it helps the researcher to “[n]ever waste time looking for and evaluating research again” (Scite 2025). It’s baffling that these arguments are successful when aimed at scientists and their supporting institutions, people that should arguably not be overly preoccupied with ease and increasing productivity in a reductionist sense. I’ll return to this, and will note for now that trying to sell AI solutions for research this way epitomizes how techno-fixes and techno-solutionism often entails reframing non-problems as problems amenable to technological solutions. While techno-fixes tend to entail identifying technological “band-aids” that allows us to skip social change, techno-solutionism is a much more ambitious endeavor aimed at using technology to radically reshape, restructure, and improve the state of things and solve a problem by addressing its more fundamental causes (Sætra and Selinger 2024). Doing research is viewed by many as a deeply valuable endeavor, but then we also have techno-fixers equipped with AI telling us that they can *relieve us* of these tasks.

But what are the tasks AI evangelists are now proposing that we can and/or should wholly or partially automate with Gen AI? Fig. 2 shows some typical steps in the research process, from the conception of an idea to its publication and dissemination.

Regardless of one’s normative evaluation of Gen AI systems, they have become capable enough to warrant the suggestion that *all* stages of the process can now be fully or partially automated or optimized with AI. My suggestion

Fig. 2 The research process



is that by doing this, we arrive at the *Research Automaton* – the machine capable of producing research like output of quality that is at least of sufficient quality to be comparable to the much of the scholarly work that has been published in various journals already, and that it is likely to surpass much of it. By saying this I do not, however, say that the quality of research will or can be *great*, or that it comes without harms, as we will soon return to.

Table 2 summarizes how Gen AI *might* be used to optimize or automate the steps in the process.

4 Discussion

Having posited that the Research Automation is now a real possibility, what remains is to consider how we should deal with this potentiality. What are the reasons to embrace the *human* side of science and research, and what are the potential reasons for embracing the automaton and turn ourselves into research *cyborgs*?

4.1 Science and academia

The nature of the scientific process, or doing science, is comprehensively explored in various treatises on, for example, the history or sociology of science. Max Weber, for instance, discussed science as a *vocation* in a 1917 lecture (Weber 1946). The title of the lecture suggests that science is more than some technical description or recipe for doing science, and it could be seen as a way of life, or a calling. If so, changing the scientific process through technological mediation also entails changing the vocation, and *life* for those doing it. Moreover, it raises the question of whether machines *can* do science, at least if they can do so without meaningful and significant human input.

In the 100 plus years since Weber explored the vocation, much has happened to many of the norms, institutions, and tools related to scientific activity. However, academia has in certain respects always been, and tried to remain, an anomaly that does not automatically adopt or adapt to the trends of society at large. As an ideal, and partially in practice still, academia adheres to traditional academic values quite at odds with, for example, the New Public Management reforms that has swept across much of the public sector (Lorenz 2014; Santiago and Carvalho 2008). Ideals of academic freedom and ideals related to academic institutions go further back, as the Humboldtian model of higher education, for example, emerged in the early nineteenth century. Here, universities are seen as arenas for the “*Bildung*”, not just professional training, of young people, etc., and these ideals still hold out in many people, and some institutions. Academia is arguably a key social institution that enables

crucial social critique, novel thinking, and innovation across a large number of domains required for flourishing societies.

4.2 Science as a normative activity

What does it entail to treat science as a vocation, as a calling or special way of life? And how is it now impacted by Gen AI? Weber (1946) referred to something more than machine-like principles when discussing the nature and role of science and the people doing it. He believed, for example, that people must have an *experience* of science to do it well. Having a personal connection to a certain scientific field was important, he argued, for enjoying it and doing it well. Finally, and perhaps most controversially, he likened the work of a scientist to that of an *artist*, a move that is quite interesting for exploring the potential limitations of the Research Automaton.

Creativity has traditionally been understood as a cornerstone of scientific activity (Sætra 2018). According to Polanyi (1962), scientific discovery is inherently creative, dependent upon intellectual passion, intuition, and heuristic insight—qualities some conceive of as deeply human and beyond the reach of even our most sophisticated machines. Similarly, Weber (1946) emphasizes inspiration and intuition as essential, stating that science without human-driven ideas risks devolving into sterile calculation. Popper (1989) echoes this view, describing scientists as driven by intellectual curiosity and passion akin to artistic creativity. These perspectives suggest that human creativity, especially the transformative type that reshapes entire fields, is an essential aspect of science. It is, importantly, an aspect that can be experienced as immensely valuable and enjoyable for humans, something that gives joy and meaning to those who are able to do this as part of their work, which makes it valuable both for these individuals and potentially for society more generally. In the innovation study on AI for materials discovery references above, the increase in innovation was accompanied by 82% of the scientists asked reporting *reduced* job satisfaction as they felt they could not be as creative as before and that their skills were underutilized (Toner-Rodgers 2024). In addition, the centrality of human creativity in many discussions of science also points to the potential for Gen AI-automated science to lack certain qualities inherent in traditional research, something I return to in the next section on the more pragmatic reasons to resist the automaton.

Science is also a social undertaking (Sætra 2018). A communal one, even. Individuals pursue knowledge, building on the work of others, standing on the shoulders of giants, and continually interacting with them. Merton’s norms for science, for example, refer to aspects such as communalism and the open sharing of results (Merton 1973). There is an inherently democratic quality embodied in science as it is

Table 2 The steps in the research process and article suggesting how AI can potentially be of use in each step

Research step	How Gen AI can be used	Examples of known Gen AI tools	Selected key references
Ideation	Idea generation and initial brainstorming; synthesis of new research themes based on existing knowledge	ChatGPT, Gemini, Claude, Mistral, DeepSeek	(Gu and Krenn 2024; Stremersch 2024)
Identifying research questions	Suggestion and refinement of research questions, based on either a perceived academic, societal, or industry need or automated identification of gaps in literature	ChatGPT with plugins, ScholarAI, Elicit, Perplexity	(Elicit 2025; Lahat et al. 2023)
Literature review	Rapid synthesis and summarization of large bodies of literature, automated extraction of key points, trends, and contradictions	Elicit, ChatGPT DeepResearch, scite_, ResearchRabbit	(Bolanos et al. 2024; Elicit 2025; OpenAI 2025)
Theoretical and methodological approach	Automated recommendation of suitable theoretical frameworks and methodological approaches based on research objectives and disciplinary norms;	ChatGPT, Claude, Gemini, Mistral, DeepSeek, Elicit	(Chubb et al. 2022; Elicit 2025)
Data gathering	Automated scraping, aggregation, and processing of textual and numerical data from various sources; interviewing and other forms of surveying humans to gather data	Diffbot, GPTs with browser plugins, ChatGPT Advanced Data Analysis	(Burleigh and Wilson 2024; OpenAI 2025)
Transcription and data handling	Automated transcription of audio and video data; quantitative data wrangling	OpenAI Whisper, Otter.ai, NVivo with AI-assist	(Kim et al. 2024; Radford et al. 2023)
Analyzing data	AI-driven statistical, qualitative, and textual analysis; AI-assisted tagging and coding of qualitative data; pattern recognition, sentiment analysis, thematic analysis;	ChatGPT Advanced Data Analysis, Claude, Gemini, Mistral, DeepSeek	(Pattyn 2024; Schwarz 2025)
The writing process	Drafting entire sections of manuscripts; paraphrasing and summarizing research findings; generating references and bibliography	ChatGPT, Gemini, Claude, Mistral, DeepSeek, GrammarlyGO	(Barrett and Pack 2023; Chartier-Edwards et al. 2024)
Peer review	Automated pre-submission manuscript assessment; performing peer review, including identification of methodological issues, paper strengths and weaknesses, and proposing decisions; revision of manuscripts and writing response to reviewers	ChatGPT, Gemini, Claude, Mistral, DeepSeek, scite_, Paperpal, Turnitin	(Gendron et al. 2022; Mehta et al. 2024)
Editorial work	Automated editing for language, style, consistency, checking manuscript adherence to journal guidelines; plagiarism and authenticity checks	GrammarlyGO, Writefull, Paperpal	(Gendron et al. 2022; Xames and Shefa 2023)
Dissemination	Generating press releases, plain-language summaries, social media content; automated translation into multiple languages for wider reach	ChatGPT, Gemini, Claude, Mistral, DeepSeek, DeepL, Canva AI	(Dwivedi et al. 2023)

both a way to actively engage with ones fellow citizens and a way to actively contribute to change in one's own society (Sætra 2018). Humans are deeply and radically social, and we might derive great joy and value from the social interactions that characterize traditional human-led research.

Merton refers to scientists as deeply embedded and integrated in their societies, and that science should not been seen as a “self-validating enterprise which was in society but not of it” (Merton 1973). Science is, at least in modern western societies, one of the key institutions for making sense of the world, generating new knowledge, deciding on what knowledge is, and resolving various forms of epistemic disagreements. For better or worse, you might say, but still: Science as an activity relates to the formation and shaping of our societies worldviews, values, and direction. That makes science a deeply normative activity, one where we might want to ensure that *humans* remain the central actor, even if that means taking a loss in terms of effectivity and productivity.

The interaction between science and society is also seen in the dynamic social construct Merton refers to as the *ethos of science*. These are the norms and values that consist of the “prescriptions, proscriptions, preferences, and permissions” that apply to scientists (Merton 1973). While this ethos is partly developed and promulgated by scientists themselves, they are also deeply shaped by society more broadly, both the public and policymakers. For Merton, the *ethos of science* is not a technical thing, but pertains to deeply moral questions, questions of “right and good” (Merton 1973). One key question, then, is whether we believe that the Research Automaton will be able to work from the basis of such an ethos, or whether the ethos will necessarily change or wither as we move from human scientists to researchers that are little more than *Gen AI jockeys* relying heavily on AI tools to do research. While their most important competency is the effective control of the technology, they risk losing touch with the field of research they're seemingly working in.

From a normative perspective, research steps such as ideation, identifying research questions, and theoretical and methodological framing, are particularly critical. These stages of the research process reflect values, societal needs, and ethical considerations, shaping not only the nature but also the purpose of research itself. Furthermore, through the process of reviewing literature, interacting with people and the world as we gather and try to make sense of data, reviewing others and being reviewed by them, and framing and writing up our research, we are involved in deeply social, communicative, and political tasks. While automating them to some degree might be possible, doing so would drastically change the researcher's contact with their societies and their control over the framing and impacts of their research.

In contrast, more mechanical tasks, like formatting references or language editing, might carry less normative

weight, suggesting they could be more safely automated without compromising the integrity of scientific practice. However, we must always remain vigilant about how the use of Gen AI in, for example, the analysis, writing, and reviewing of research will introduce and manifest values built into the systems we use. Gen AI, as other technologies, is *political* (Winner 1980), and even if some use-cases might be less risky than others, the technology will have social and political implications.

4.3 Science as the pragmatic pursuit of objective excellence

The implicit question motivating this article is: Do we really need humans to do science? The death of many a profession has been prophesized with a nod to AI in recent years, but most of them are still alive and kicking. Perhaps most famous are the pesky radiologists that were supposed to be gone by now according to Geoffrey Hinton, one of AI's many purported “godfathers” (Alvarado 2022).

Moving from ideas related to the intrinsic value of having humans do research, we can also offer a radically different interpretation of the Mertonian norms of science (Merton 1973), one that could lead us to see the emergence of the Research Automaton as a good thing. The norms refer to openness, disinterestedness, universalism, and organized skepticism, in theory, norms the automaton might adhere to or could be aligned with.

For example, AI-driven research might enhance objectivity, particularly mechanical and aperspectival objectivity (Daston 1992). Machines, devoid of personal biases and individual perspectives, can potentially achieve a form of objective consistency difficult for human researchers to maintain. This form of objectivity, however, might inadvertently neglect the nuanced contextualization and ethical judgment that human researchers provide (Sætra 2018).

Thus, although objectivity in the mechanical sense *could* improve through automation, it remains insufficient without human oversight to manage interpretation, meaning-making, and ethical implications. The data and models we rely on are *value-laden* and characterized by the countless choices and theoretical and cultural leanings of the humans that have worked to collect and categorize them (Sætra 2018; Sayer 1992). If this is true, machines cannot be truly objective in a less mechanic sense, which again means that whenever we try to outsource such tasks to Gen AI, we also outsource decisions related to values and morality, without retaining much control over the efforts of such decisions and *which* morals and values are then relied upon.

Another major danger of automating the scientific process is the potential atrophy of critical thinking skills essential for any kind of critique, innovation, and human development. If scientists stop engaging deeply in tasks, such as literature

reviews, theoretical reflection, and data interpretation, critical thinking and deep comprehension skills may diminish over time (Sætra 2018, 2023, 2025). This potential is currently being discussed as risks related to “cognitive offloading” as many hypothesize that Gen AI can boost productivity, but at a cost in terms of human learning and cognitive development (Nosta 2025).

If we believe that practice makes perfect, there are good reasons to fear that not practicing reading complex and long texts, not summarizing, analyzing, and contextualizing them, and not writing and translating our own thoughts into communications aimed at different groups for specific purposes, will over time make us *worse* at those tasks. While we may gain in effectiveness, and our expertise in using Gen AI systems might improve, the loss of the aforementioned cognitive skills could spell grave danger for science and research as a human endeavor (Abbas et al. 2024).

While we might not yet have sufficient evidence base to authoritatively state the short-, medium-, and long-term consequences of Gen AI use, some tentative results *do* suggest that, for example, the use of such systems negatively impacts critical and creative thinking (Lee et al. 2025; Toma and Yáñez-Pérez 2024). And when we consider the risks involved, it might make sense to adopt a precautionary principle rather than to consider controlled quantitative studies as the only basis for responding to the Research Automaton.

Moreover, automation might encourage passive acceptance of findings, reducing critical skepticism (Merton 1973) and weakening organized scientific critique. This passive stance risks undermining the collective self-corrective mechanism essential to scientific progress. We can also envision a form of scientific-cultural atrophy resulting from overreliance on automated tools and outsourcing to machines of cognitively taxing, and stimulating, tasks (Pomeroy 2025; Sætra 2018). Scientific communities risk losing shared expertise, tacit knowledge, and the ability to innovate independently of algorithmic guidance, and such a loss could eventually also constrain the very AI systems relied upon as these systems depend fundamentally on high-quality human-generated insights and continuous innovation to update and improve their models.

Could we find ourselves in a situation in which human cognitive and scientific-cultural atrophy makes humans increasingly unable to produce valuable new knowledge? This could be accompanied by technological development and increased *machine* learning, however, which could provide us with an initial increase in scientific output. However, the loss of *human* learning could then gradually negatively impact our capacities for developing new knowledge and better AI models as we lose the very input that has enabled our machines to grow ever more adept, through what is referred to as “model collapse” from training new models on synthetic (Gen AI-generated) data (Shumailov et al. 2024).

Doing science is arguably crucial for human learning and community development (Sætra 2018). For individuals, the hands-on process of doing science builds cognitive capacities, critical reasoning skills, and deep conceptual understanding. For communities, active participation in science fosters collective intellectual growth, societal cohesion, and adaptive capacity. Deeply automating the science process could deprive us of vital learning arenas and experiences, impairing long-term knowledge development and innovation.

4.4 Beyond the automaton: augmentation and hybridity

While the preceding discussion highlighted significant normative and pragmatic reasons to resist the wholesale automation embodied by the Research Automaton, this should not be read as a complete rejection of AI in science. The critical question is: How and when can Gen AI be integrated in research in ways that *augment* human capabilities and enrich the research *process*, rather than merely optimizing for *product* output at the expense of human development and engagement? If the intrinsic value lies in the process, can the process *evolve* into something even better and more meaningful through human–AI interaction?

Viewing AI not as a replacement but as a potential cognitive *scaffolding* (Sætra 2022) or prosthetic (Hayles 1999) opens possibilities for *augmentation*. These perspectives suggest a spectrum of AI integrations across the steps in the research lifecycle, with varying implications for process value.

For instance, AI tools might automate or significantly speed up tasks often considered more mechanical or less central to core intellectual discovery. Many would here probably think of transcription and data handling tasks, perhaps initial literature reviews, editorial work and proofreading, and various dissemination tasks. Using AI here could certainly free up valuable human time and cognitive resources, potentially allowing researchers to engage more deeply in the more demanding stages ideation, development of research questions, identifying and applying relevant theories and methods, getting to grips with the data in empirical work, and then deciding and executing on the framing and presentation of the research.

However, the potential for augmentation clearly also extends into these core cognitive stages. Perhaps this is where there is simultaneously both the most positive potential *and* the most severe risks. Could AI, used critically, act as a brainstorming partner during ideation, suggesting novel connections or overlooked perspectives (Gu and Krenn 2024)? Could it assist in identifying research questions by rapidly mapping literature gaps (Lahat et al. 2023) and identifying research needs and strategic research opportunities,

prompting deeper human reflection? In analyzing data, AI might reveal complex patterns beyond human perception (Edwards 2010), acting as an epistemological instrument that reshapes understanding. Even in the writing process, AI might assist in structuring arguments or overcoming writer's block, provided the human retains intellectual control. In such a case, AI could act as the more competent *other* as theorized by the socio-cultural learning theorist Vygotsky, an entity that “scaffolds” human learning by operating in the human’s *zone of proximal development*, dragging humans toward new domains of knowledge and understanding (Sætra 2022).

This suggests a model less like the automaton and more like a “cognitive assemblage” (Hayles 2017) or a form of hybrid, collective intelligence (Hui, 2024). In such a model, human researchers leverage AI’s processing power while maintaining meaningful control over the crucial elements of critical judgment, ethical evaluations, contextual understanding, reflexivity, and the understanding and guiding of the social impacts of research. The research *process* itself would then transform into one of human–AI interaction, supervision, and interpretation. Meaning and value might be found not only in the final output but also in the skilled navigation of this hybrid workflow. The potential for AI systems, with their “inhuman” ways of processing information (Fazi 2021; Parisi 2019), to spark genuinely new forms of human imagination and break through existing knowledge barriers should not be dismissed lightly. There is clearly a potential for Gen AI to be involved in even the creative aspects Weber (1946) and Polanyi (1962) valued so dearly.

Yet, even this more optimistic vision of augmentation is not without risks. The dangers of deskilling remain if humans become passive supervisors rather than active intellectual partners. The complexities of tool mediation (Kaptelinin and Nardi 1997) mean that integrating AI effectively requires new skills and critical awareness. Automation bias, the tendency to over-trust machine outputs (Mosier and Skitka 1999), could still undermine critical skepticism. Furthermore, the reconfiguration of research work could exacerbate existing inequalities or create new dependencies (Danaher 2019). Dependencies both on machines and systems themselves, but not least on the people that develop and deploy them, a crucial question to consider if we’re taking the current world of Big Tech dominance as the point of origin.

Therefore, while rejecting the simplistic efficiency narrative of the Research Automaton, we must consider the potential for *managed augmentation*. The key lies in consciously directing AI toward supporting, rather than supplanting, the core human activities that give the research process its normative value through its social, democratic, epistemic, and developmental consequences for individuals and society. One challenge will then be to differentiate

between automating drudgery (potentially beneficial) and automating thought (potentially detrimental), ensuring that technology serves to enhance, not diminish, the human capacity for meaningful scientific inquiry across all stages of the research process.

Another challenge will be to decide whether we *want* our cognitive structures to be infiltrated by AI systems and the people and corporations that develop them, and *why* if so. Some might see such structures as the key to further human growth and flourishing, others will see such a move as one motivated by the instrumental considerations of how these structures might give us increased efficiency.

5 Implications for practice

The emergence of Gen AI necessitates a proactive and critical response from all stakeholders within and beyond the scientific ecosystem. I now translate the preceding considerations into tentative implications for practice, formulated as my subjective advice to the key stakeholders involved. This also means that the implications here presented should be taken as the starting point for discussion, as others’ analyses and evaluations of the discussions presented in this article most likely lead to quite different sets of advice.

The stakeholders I focus on are the following: *Early-career researchers (ECRs)* are key stakeholders as the formative stages of a research career are particularly vulnerable to the potential downsides of over-reliance on Gen AI. While these tools offer tempting shortcuts, ECRs need the fundamental skill development that occurs through direct, sometimes arduous, engagement with the research process itself. At the same time, some are worried that *not* harnessing the power of new technologies might lead to a situation where doctoral education and academia loses relevance (Jung 2022).

Supervisors play a critical role in shaping the next generation of researchers and navigating the integration of Gen AI into academic practice, and their guidance can either accelerate detrimental trends or foster responsible engagement with new technologies. Supervisors are also *researchers*, and in this role they are not immune to the pressures and temptations of automation.

Academic institutions shape the environment in which research occurs and have a significant role in setting norms and providing resources, and *policymakers* govern research policies, funding, and assessment, and through this they have significant influence on the research landscape. Finally, *publishers* are relevant as key gatekeepers and disseminators of scientific knowledge, facing unique challenges and responsibilities from the Research Automaton.

5.1 Early-career researchers

Dive into deep academic traditions and theories and seek deep knowledge. Traditional education entails being “raised” in one or a few academic disciplines, and through that process to become deeply immersed in specific sets of theories or approaches to the world. This occurs through deep engagement over time. Often with long and arduous texts and sources, sources that will be further and further removed from the short context windows modern individuals tend to engage with on, for example, social media platforms like X or TikTok. Learning to struggle through such sources builds a deep understanding and knowledge of a field that will likely not be gained through more superficial summaries or AI-based mediations of academic literature.

Prioritize long-term skill development over speed by embracing growth through struggle. Understand that wrestling with complex literature, manually coding qualitative data, debugging code, and structuring a difficult argument are often not meaningless obstacles to be overcome, but are the very activities that build deep understanding, critical thinking, methodological rigor, and creativity. Resist the urge to outsource these core intellectual tasks entirely to AI, even if it means a slower pace initially. The manual and iterative process of literature review, data analysis, and writing provides invaluable tacit knowledge and formative experiences essential for developing independent research expertise. Automating these steps prematurely risks creating a superficial understanding and hindering the development of robust, transferable research skills. Your future ability to push boundaries and generate genuinely novel and important ideas depends on the foundational skills built today.

Adopt hybrid approaches cautiously and strategically. Learn to use AI tools judiciously, but don’t prioritize Gen AI skills, such as prompt engineering, over the more foundational research and cognitive skills discussed above. Gradually start experimenting with Gen AI for tasks where they genuinely augment your capabilities without replacing core intellectual effort; for example, using transcription services for interviews (while still engaging deeply with the content), employing AI for grammar checks and formatting, or using it as a brainstorming partner whose suggestions are critically evaluated, not blindly accepted. As a developing researcher, strive to always develop your first drafts independently of Gen AI, and start using these systems as reviewers and feedback machines. Always reflect on how and why you are using a tool in each instance, and what it does to your research, and you.

5.2 Academics and others with a supervisor hat

Provide active mentorship on AI use. From the get-go, start any supervision relationship with a thorough discussion of

your and the candidate’s stances on Gen AI use in research. What is acceptable (to you both) and what sort of “contract” do you agree on regarding the candidate’s use of Gen AI in their work? This dialog should ideally start before the formal supervision relationship begins as it can be part of position announcements and interview processes. I personally, for example, would not now hire a candidate who could not or would not be willing to start their research training process without the aid of Gen AI tools. While not necessarily straight-out banning the use, engage in explicit discussions with ECRs about the ethical and pedagogical implications of using Gen AI. Guide them in making informed choices about *when* AI might be helpful and *when* manual effort is crucial for their intellectual growth and skill development. Teach them to rigorously verify AI suggestions, understand the limitations and potential biases of these tools, and maintain intellectual ownership over their work. Finally, *don’t* rely on Gen AI in your supervision and create a situation in which you become little more than what the candidate could get from these systems on their own.

Champion skill preservation and process Value. In mentoring and assessment, consciously prioritize the development of critical thinking, creativity, methodological understanding, and intellectual independence over productivity or output volume. Reinforce the idea that the *process* of becoming a researcher is as important as the papers produced, and stress the idea that you need to see their own independent, and imperfect, work to learn to know and supervise them.

Uphold ethical and scientific responsibility. Supervisors embody and transmit the ethos of science, and have an ethical responsibility to foster research norms grounded in integrity, critical inquiry, and intellectual honesty. Do your best to push back against purely instrumental or efficiency-driven motivations encouraged by AI salespeople and evangelists and institutional pressures, to allow for the slower and deeper development of your ECRs.

5.3 Academics and others with a researcher hat

Discuss Gen AI use in collaborations. As for supervisors, researchers must now also be *very* aware of the fact that their collaborators are likely using Gen AI in a myriad of ways in their various cooperative projects. It is imperative these days to discuss what is acceptable and desirable, what isn’t; for example, what can be uploaded to what service, what sort of revisions can be introduced based on a Gen AI assessment?

Guard creativity, originality, and quality. Reflect critically on how delegating tasks like literature synthesis, outlining, or drafting to AI might affect the depth, originality, and serendipity of your own thinking. Gen AI use across the research process will inevitably shape and guide the developmental trajectory of yourself and your research. Always strive to understand how, and always avoid usage

levels that lead to a feeling of loss of both control over this trajectory and ownership of your research. Uphold the highest standards of scientific validity and quality. AI-generated text, code, or analysis must be treated with caution, and the ultimate responsibility for the work's integrity remains with the human researcher.

Explore and document “Hybrid Intelligence”. Experiment with human–AI collaborations that enhance research quality and insights, not just speed. If for no other reason than to understand the limitations and capabilities of the technology, document and share successful models of hybrid workflows where AI augments human capabilities in meaningful ways, contributing to a community understanding of best practices.

Pioneer and develop new forms of dissemination. If you value human research and researchers, devote time and energy to arenas and developmental projects that may lead to new forms of academic interactions and engagement. Journal publications, for example, are liable to be severely disrupted and infested with Gen AI, whereas conferences and workshops will tend to retain more of the significant human engagement with the topics and materials presented and discussed. When reviewing and editing journals, you also have a chance to enforce the ethos of science *you* stand for.

5.4 Research and academic institutions

Critically evaluate AI tools and vendors. Resist the hype often presented by vendors. Evaluate potential institutional investments in Gen AI tools based on their alignment with core educational and research values (promoting critical thinking, research integrity, equity) rather than solely on promises of increased productivity or efficiency savings. Be extremely wary of tools marketed as “fixes” for the “problem” of doing research.

Develop clear guidelines and robust training. Establish clear institutional policies regarding the ethical and acceptable use of Gen AI in research and teaching, covering issues like transparency, plagiarism, and data privacy. More importantly, however, state how you conceptualize *research* and its value, and the positive role of humans in it. Complement these guidelines with training programs that foster critical AI literacy among both staff and students. However, always remember that deep disciplinary immersion is both more important and more difficult to attain than mastery of the latest and greatest AI system.

Foster a culture of integrity and quality. Actively promote a research culture that values intellectual depth, originality, methodological rigor, and ethical conduct above sheer publication volume. This requires aligning incentive structures (e.g., promotion criteria, internal funding) with these values and explicitly discouraging practices that prioritize output over process integrity. Focus relentlessly on disseminating

knowledge about the harms and unacceptability of paper mills, citation cartels, and predatory journals and conferences, and be very sure not to reward those that rely on such methods to boost their metrics.

Invest strategically in responsible AI integration. Allocate resources not just for AI tools themselves, but also for the infrastructure, training, and support needed to foster responsible, critical, and sustainable use of AI in hybrid research models. This entails building AI, not just Gen AI, expertise in-house, and developing and serving local Gen AI systems that reduce the reliance on Big Tech providers.

5.5 Policymakers

Develop a long-term vision for research. Recognize that prioritizing short-term efficiency gains through automation may jeopardize the long-term viability and innovative capacity of national research systems, if it leads to widespread skill atrophy. Incentivize educational and research practices that preserve and cultivate essential research skills. These are skills related to critical thinking and general skills that all students and researchers will require to succeed in whatever future we realize.

Reform research assessment frameworks. Critically review and adapt national research assessment exercises and funding criteria. Move away from overly simplistic, quantitative metrics that incentivize volume and speed, which Gen AI automation can easily inflate. Instead, develop and implement assessment methods that reward substantive intellectual contributions, originality, methodological soundness, reproducibility, and the societal relevance of research, qualities that are harder to automate and reflect the true value of the scientific process.

Promote ethical oversight and governance. Support the development and implementation of ethical frameworks and governance structures for the use of AI in research, addressing issues, such as algorithmic bias, data privacy, intellectual property, and accountability. Develop and enforce clear policies regarding the transparent and ethical use of Gen AI in publicly funded research. Ensure that grant proposals and reports accurately disclose the role of Gen AI tools.

5.6 Publishers

Stay strong and stand for something. First and foremost, it's imperative that publishers don't fall for the narrative that AI research is inevitable and that they must simply jump on this train to stay relevant. Take a stance on what science is to you, and *don't* fall for the temptation to turn all the contributors that spend their efforts to provide you with content and reviews into suppliers of novel data for AI training models.

Establish clear editorial standards and authorship policies for AI Use. Communicate explicit policies regarding the

acceptable use and necessary disclosure of Gen AI tools in manuscript preparation. Without reliable detection methods, however, this point is as of now largely unenforceable. Reiterate and enforce clear authorship criteria stipulating that only humans can be authors. Provide unambiguous guidelines on how the use of Gen AI tools should be appropriately acknowledged in manuscripts, differentiating tool use from intellectual contribution.

Invest in quality control and integrity mechanisms. Develop and implement robust mechanisms to detect and evaluate AI-generated or AI-manipulated submissions, including potentially fabricated data or plagiarized text. This may involve using AI detection tools (while acknowledging their limitations) and refining editorial screening processes to uphold scholarly standards against threats like sophisticated paper mills (Prez-Neri et al. 2022). Invest in the development of *new* modes of detecting and monitoring Gen AI use to overcome current shortages, but always remember that automatic screening tools are never neutral or reliable, but normative governance instruments (Hesselmann 2023). While AI might offer tools to assist peer reviewers (e.g., identifying relevant literature, checking for plagiarism, assessing reporting standards), the core evaluative judgment must remain human. Ensure editorial decisions are transparent, accountable, and made by humans.

6 Conclusion

In the vignette, I depicted a future where the scientific paper, as a product, could be generated with minimal human engagement. I've argued that such a future, driven by the capabilities of Gen AI and framed by narratives of efficiency and techno-solutionism, poses a clear threat to the core values of science. The central argument is that the value of scientific inquiry lies not merely in its outputs, papers, patents, and products, but in the *process* itself. This process encompasses experiences of intellectual struggle, critical engagement with complex ideas, creative problem-solving, collaborative discovery, and becoming enmeshed in an academic tradition steeped with both explicit and tacit knowledge.

The rise of the *Research Automaton*, whether as a fully autonomous entity or as a suite of tools that radically automates core intellectual tasks, challenges this process-centric view. I have identified both normative and pragmatic reasons to resist its unchecked proliferation. Normatively, viewing science as a vocation highlights its role in fostering intellectual passion, ethical reflection, and democratic engagement, aspects potentially diminished when human judgment and experience are sidelined. Pragmatically, the outsourcing of cognitive effort to machines risks the atrophy of essential research skills, particularly critical thinking and creativity, potentially leading to a long-term decline in scientific

innovation even if short-term productivity metrics increase. Furthermore, the embedded values and biases within AI systems always raise concerns about objectivity and the perpetuation of existing inequalities.

This is not to deny the potential benefits of AI in science. The vision of hybrid intelligence, of human researchers working with AI partners within cognitive assemblages, holds some promise. However, realizing this positive potential requires navigating a crucial distinction: are we using AI to enhance the human-driven process of discovery, or are we allowing it to replace that process in pursuit of optimized output? The powerful narrative of techno-solutionism, often wielded by technology vendors and embraced within institutions facing efficiency pressures, actively blurs this line, reframing the valuable effort of research as an inefficiency to be eliminated.

The path forward will be shaped by the conscious choices made by researchers, educators, institutions, policymakers, and publishers. As outlined in the implications section, navigating this era demands critical evaluation of AI tools, thoughtful adaptation of pedagogical approaches and assessment metrics, a renewed commitment to research integrity, and the fostering of a research culture that values critical thinking and the development of human cognitive capabilities.

The key question is not simply *can* we automate science, but *should* we? We must ask ourselves: What kind of scientific future do we want to build? And what role can and should Gen AI have in ensuring that science provides individuals and society with meaningful lives and the knowledge we require to reach our goals?

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